

Qian Hu

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EDUCATION & RESEARCH

Australian National University

2025.12 - Present **Visiting PhD student** Supervised by Daniel Noble
Project: Thermal plasticity of avian embryonic development

Beijing Forestry University

2021.09 - Present **PhD** Supervised by Jianqiang Li & Jiliang Xu
Project: Temperature-driven reproductive strategies, embryonic development, and adaptive evolution in uniparentally and biparentally incubating birds

2017.09 – 2021.06 **Master** Supervised by Jianqiang Li
Project: A comparative study of different incubation modes and their adaptability of two *Aegithalos* species

2013.09 – 2017.06 **Bachelor** Supervised by Gangming Zhang
Project: Phylogenetic analysis of the Asian “*Cheilanthes*” — Based on two nuclear DNA sequences

RESEARCH FOCUS

- **Avian Reproduction & Parental Care**

Studying how birds regulate reproduction through reproductive strategies, parental investment, and incubation behaviour, and how environmental conditions shape parental care strategies.

- **Incubation Behaviour & Thermal Ecology**

Studying how birds regulate the thermal environment of developing embryos through incubation behaviour, and how temperature influences incubation investment, incubation strategies, and reproduction.

- **Embryonic Development & Developmental Plasticity**

Studying how temperature and other early-life environmental conditions shape embryonic development, phenotypes, and survival, and how developmental responses contribute to adaptation to environmental change.

- **Comparative Physiology & Evolutionary Ecology**

Using comparative and evolutionary approaches to investigate how physiological and behavioural responses to environmental variation are shaped by reproductive strategies, ecological conditions, and evolutionary history.

SKILLS

- **Field and lab skills:** Nest searching & monitoring, Breeding behaviour monitoring, Bird banding, Wildlife photography, PCR, Embryonic heart-rate monitoring (Buddy), Songbird blood sampling for hormone analysis.
- **Computer skills:** Windows, R (*lme4*, *MuMIn*, *RMark*, *MCMCglmm*, *ggplot2*, *metafor*, *orchaRd*, etc.), SPSS, Microsoft Office, Adobe Creative Suite (Photoshop, Illustrator, Premiere).
- **Language:** Mandarin, English.

PUBLICATIONS

First author

1. **Hu, Q.**, Wang, Y., Wang, H., Wang, Y., Xia, X., Xu, J., & Li, J. (2026). Parental care and bird embryonic metabolism: A comparison between uniparentally incubating Silver-throated Tits and biparentally incubating Black-throated Tits. *Functional Ecology*, 40(4), 800-815.
2. **Hu, Q.**, Xu, J., & Li, J. (2026). Ecological and Life-History Correlates of Incubation Attentiveness Differ Between Female-Only and Biparentally Incubating Passerine Birds. *Global Ecology and Biogeography*, 35(3), e70225.
3. **Hu, Q.**, Wang, Y., Yu, G., Lv, L., Wang, P., Wen, Y., ... & Li, J. (2025). Biparentally incubating *Aegithalos concinnus* (Black-throated Tit) provide a more stable nest thermal environment than uniparentally incubating *A. glaucogularis* (Silver-throated Tit). *Ornithology*, 142(4), ukaf025.
4. **Hu, Q.**, Wang, Y., Yu, G., Lv, L., Wang, P., Wen, Y., ... & Li, J. (2024). The effect of ambient temperature on bird embryonic development: a comparison between uniparental incubating silver-throated tits and biparental incubating black-throated tits. *Journal of Avian Biology*, 2024(5-6), e03168.
5. **Hu, Q.**, Wen, Y., Lv, L., Wang, P., Li, J., & Xu, J. (2023). Does research activity affect nest survival of birds? A case study on the black-throated tit (*Aegithalos concinnus*). *Biodiversity Science*, 31(6), 22570. [in Chinese] 胡骞, 文野, 吕磊, 王鹏程, 李建强, & 徐基良. (2023). 研究活动是否影响鸟类的巢存活率?——以红头长尾山雀为例. *生物多样性*, 31(6), 22570.
6. **Hu, Q.**, Wen, Y., Yu, G., Yin, J., Guan, H., Lv, L., ... & Li, J. (2020). Research activity does not affect nest predation rates of the Silver-throated Tit, a passerine bird building domed nests. *Avian Research*, 11(1), 28.
7. **Hu, Q.**, Zang S., Xu K., Xu L., Wang X., Ding C., & Li J. (2019). Bridled Tern *Onychoprion anaethetus* and Aleutian Tern *O. aleuticus* Found in Shandong Province. *Chinese Journal of Zoology*, 54(3), 612-613. [in Chinese] 胡骞, 臧少平, 徐克阳, 薛琳, 王希明, 丁长青, & 李建强. (2019). 山东发现褐翅燕鸥和白腰燕鸥. *动物学杂志*, 54(3), 612-613.

Co-author

8. Wang, H., **Hu, Q.**, Wang, Y., Lv, L., Wang, P., Zhang, Z., & Li, J. (2026). Density and kinship interactively affect natal dispersal in a social bird. *Proceedings of the Royal Society B: Biological Sciences*, 293(2070), 20252985.
9. Li H., Liu C., Huang X., Li T., **Hu Q.**, Wang Y., Li J., Yan L., & Zhang D. (2026). The mitochondrial genome of the *Ornithophila metallica* (Diptera: Hippoboscidae) and its phylogenetic significance. *Entomotaxonomia*, 48(1), 58-70. [in Chinese] 李寒晴, 刘纯焱, 黄晓芳, 李淳滢, 胡骞, 王悦, 李建强, 闫利平, & 张东. (2026). 金光喜鸟虱蝇线粒体基因组及系统发育分析(双翅目:虱蝇科). *昆虫分类学报*, 48(1), 58-70.
10. Li, X., Liu, Z., **Hu, Q.**, Peng, Y., Wang, H., Li, J., ... & Xu, J. (2025). Preliminary study of the diversity of birds in Turpan-Hami Basin. *Arid Land Geography*, 48(10), 1707-1720. [in Chinese] 李新宇, 刘正霄, 胡骞, 彭于杨, 王辉, 李建强, ... & 徐基良. (2025). 吐哈盆地鸟类多样性的初步研究. *干旱区地理*, 48(10), 1707-1720.
11. Peng, P., Chen, W., Qianhan, N., Yu, H., **Hu, Q.**, Zhu, J., ... & Li, J. (2025). Hanging Position of Artificial Nest Boxes Affects Reproductive Success of Russet Sparrow *Passer cinnamomeus*. *Animals*, 15(10), 1413.
12. Jin, T., **Hu, Q.**, Liu, Y., Li, J., Pan, S., & Xu, J. (2025). Latitudinal variations in incubation behaviour among uniparental incubating birds. *Journal of Animal Ecology*, 94(4), 800-810.
13. Wang, Y., **Hu, Q.**, Wang, Y., Liu, J., Du, Z., Xu, J., & Li, J. (2024). Selective effect of winter weather on a songbird's morphology depends on individual sex and winter condition. *Oecologia*, 205(2), 339-350.
14. Jin, T., Lu, S., Wang, Y., Hua, J., Liu, Z., **Hu, Q.**, ... & Xu, J. (2024). The clutch size, incubation behavior of Reeves's Pheasant (*Syrnaticus reevesii*) and their responses to ambient temperature and precipitation. *Avian Research*, 15, 100168.
15. Wang, Y., **Hu, Q.**, Xu, J., & Li, J. (2023). Sex-specific selective effect of winter weather on morphological traits in a small passerine bird. *Avian Research*, 14, 100093.
16. Zhu J., Huang T., Du Z., **Hu Q.**, & Li J. (2023). Nesting Plant Affects the Nest Predation Risk of the Black-Throated

- Tit *Aegithalos concinnus*. **Chinese Journal of Zoology**, 58(3), 330-340. [in Chinese] 朱家贵, 黄涛, 杜志勇, 胡骞, & 李建强. (2023). 营巢植物影响红头长尾山雀的巢被捕食风险. *动物学杂志*, 58(3), 330-340.
17. Lu, S., Liu, Z., Tian, S., Song, K., Hu, Q., Li, J., & Xu, J. (2022). Sex-specific movement responses of Reeves's pheasant to human disturbance: Importance of body characteristics and reproductive behavior. **Animals**, 12(13), 1619.
 18. Zhang, Y., Huang, T. T., Hu, Q., & Zhu, J. N. (2022). Analysis on the habitat design strategy of community park based on the improvement of bird diversity. **Chinese Landscape Architecture**, 38(3), 106-111. [in Chinese]张颖, 黄婷婷, 胡骞, & 朱建宁. (2022). 基于鸟类多样性提升的社区公园生境 营造策略探析. *中国园林*, 38(3), 106.
 19. Yin J., Yu G., Nursoltan Q., Hu Q., Wang Y., & Li J. (2021). Sexual Size Dimorphism and Its Seasonal Variation in Cinereous Tit *Parus cinereus*. **Chinese Journal of Zoology**, 56(6), 826-835. [in Chinese] 尹江南, 余高阳, 努尔索力坦·恰汗, 胡骞, 王悦, & 李建强. (2021). 大山雀身体大小的性二态及其季节变化. *动物学杂志*, 56(6), 826-835.
 20. Shi J., Hu J., Hu Q., Jiang L., Zhao Y., Xu J. (2020). Chestnut Thrush *Turdus rubrocanus* Found in Dongzhai National Nature Reserve, Henan Province, China. **Chinese Journal of Zoology**, 55(3), 305-305. [in Chinese] 石江艳, 华俊钦, 胡骞, 蒋淼, 赵玉泽, & 徐基良. (2020). 河南董寨国家级自然保护区发现灰头鹀. *动物学杂志*, 55(3), 305-305.

HONORS & SCHOLARSHIPS

2024	One-year Visiting PhD Scholarship from the China Scholarship Council (CSC)
2022	Outstanding Student Cadre
2021	Excellent master thesis (school level)
2017	Excellent bachelor thesis (school level)
2016	National Encouragement Scholarship, China
2015	Outstanding Student Cadre

PRESENTATIONS & POSTERS

2026	Presentation	Parental care and bird embryonic metabolism: a comparison between uniparentally incubating Silver-throated Tits and biparentally incubating Black-throated Tits	20th International Society for Behavioral Ecology Congress	Turin, Italy
2023	Presentation	A comparative study on different incubation patterns and adaptability of two <i>Aegithalos</i> species	The 8th China Forestry Academic Conference	Harbin, China
2022	Presentation	The effect of ambient temperature changes on incubation strategies in two bird species with different incubation modes	The 16th Kingfisher Forum	Online
2021	Poster	The variation of eggshell patterns in two <i>Aegithalidae</i> species	2021 Beijing Zoological Society Academic Symposium	Beijing, China
2019	Poster	Nest visit frequency by research does not affect the nesting success of two domed-nest birds	The 15th China Ornithological Congress	Changchun, China

2019	Presentation	Does research activity affect nest survival of birds? A case study on two <i>Aegithalidae</i> species	2019 Beijing Zoological Society Academic Symposium	Beijing, China
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REFERENCES

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